

An exact non-free matrix extreme point of the two-variable wild disc

Solution packet for arXiv:2212.00748

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Abstract

Epperly, Evert, Helton, and Klep ask whether a real free spectrahedron in two variables can have a matrix extreme point which is not free extreme, and they “strongly strongly conjecture” this for the wild disc. We give such a point at level $n = 8$. The entries are specified exactly: one coordinate is rational, while the other is a symmetric zero of a quadratic matrix equation in a certified rational ball. Exact rational residual bounds prove existence of that zero and prove full column rank of the matrix-extreme system. A strict $15 < 16$ equation count then proves that the point is not Arveson extreme and hence not free extreme. All proof-critical inequalities are checked with integer and rational arithmetic by the accompanying script.

1 The question and the answer

Let

$$A_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

The associated wild disc is

$$\mathcal{D}_A(n) = \left\{ (X, Y) \in \text{Sym}_n(\mathbb{R})^2 : L_A(X, Y) = \begin{pmatrix} I & X & Y \\ X & I & 0 \\ Y & 0 & I \end{pmatrix} \succeq 0 \right\}.$$

Equivalently, by a Schur complement,

$$(X, Y) \in \mathcal{D}_A(n) \iff I - X^2 - Y^2 \succeq 0.$$

This free spectrahedron is bounded. Section 5.3 of the authors’ current manuscript says that the existence of non-free matrix extreme points for $g = 2$ is open and makes the specific wild-disc conjecture; see the included source PDF and the page-17 source image.

Theorem 1 (Two-variable wild-disc conjecture). *There exist real symmetric 8×8 matrices X, Y such that (X, Y) is a matrix extreme point of the wild disc but is not a free extreme point.*

The rest of the packet gives the exact construction and proof.

2 Exact data

Set $X = M/30$ and $Z = R/30$, where

$$M = \begin{pmatrix} -3 & 5 & -3 & 1 & -3 & -4 & -3 & 0 \\ 5 & -1 & 5 & -3 & -5 & -2 & 3 & -1 \\ -3 & 5 & 2 & -2 & 1 & 3 & 0 & 1 \\ 1 & -3 & -2 & 5 & -4 & 3 & -2 & 2 \\ -3 & -5 & 1 & -4 & 3 & -5 & 1 & 1 \\ -4 & -2 & 3 & 3 & -5 & 3 & -5 & -1 \\ -3 & 3 & 0 & -2 & 1 & -5 & 2 & -5 \\ 0 & -1 & 1 & 2 & 1 & -1 & -5 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 5 & 3 & -1 & 0 & 1 & 5 & -4 & 2 \\ 1 & 2 & -3 & -5 & -1 & -5 & 0 & -5 \\ 5 & -3 & 5 & 3 & -3 & 1 & -5 & 0 \end{pmatrix}.$$

Put

$$P = I - X^2 - Z^T Z.$$

We next specify a rational matrix $Q = N/10^{15}$ near an exact symmetric square root of P . To keep the integers readable, write $N = [N^{(1)} \ N^{(2)} \ N^{(3)} \ N^{(4)}]$ in two-column blocks:

$$N^{(1)} = \begin{pmatrix} 469136539517107 & 418057100896713 \\ 418057100896713 & 226044565094389 \\ -513730559248146 & 316155862074043 \\ -19705331325464 & -591487703586026 \\ 115589350876164 & -404383337428309 \\ 315461324061599 & -102525629138037 \\ -262136144216438 & 105847039317755 \\ 126120353785681 & -68076824009979 \end{pmatrix}, \quad N^{(2)} = \begin{pmatrix} -513730559248146 & -19705331325464 \\ 316155862074043 & -591487703586026 \\ 343845406695791 & -286080313659530 \\ -286080313659530 & -185003309748076 \\ -5713670391864 & -525675862063652 \\ 441714042553604 & 313713354292056 \\ -342388858747631 & -197186242248850 \\ 160943853597458 & 47691229959701 \end{pmatrix},$$

$$N^{(3)} = \begin{pmatrix} 115589350876164 & 315461324061599 \\ -404383337428309 & -102525629138037 \\ -5713670391864 & 441714042553604 \\ -525675862063652 & 313713354292056 \\ 647636604497521 & 110185471831049 \\ 110185471831049 & 546819792547518 \\ -79262335507014 & 306432223723976 \\ 209718944007 & -160736732966099 \end{pmatrix}, \quad N^{(4)} = \begin{pmatrix} -262136144216438 & 126120353785681 \\ 105847039317755 & -68076824009979 \\ -342388858747631 & 160943853597458 \\ -197186242248850 & 47691229959701 \\ -79262335507014 & 209718944007 \\ 306432223723976 & -160736732966099 \\ 721476885267441 & 110205575101764 \\ 110205575101764 & 918313788896616 \end{pmatrix}.$$

The displayed blocks show directly that Q is symmetric.

Choose a symmetric matrix Y in the certified ball

$$\|Y - Q\|_{\max} < 3.014 \cdot 10^{-12} \quad (1)$$

which satisfies $Y^2 = P$. Existence follows from Lemma 1; thus this is an exact implicit algebraic specification of Y , not a floating-point definition. The fixed point used in that lemma is unique in its smaller closed contraction ball, although uniqueness is not needed here.

For orientation only, Y begins numerically as

$$\begin{pmatrix} .469136540 & .418057101 & -.513730559 & -.019705331 & .115589351 & .315461324 & -.262136144 & .126120354 \\ .418057101 & .226044565 & .316155862 & -.591487704 & -.404383337 & -.102525629 & .105847039 & -.068076824 \\ -.513730559 & .316155862 & .343845407 & -.286080314 & -.005713670 & .441714043 & -.342388859 & .160943854 \\ -.019705331 & -.591487704 & -.286080314 & -.185003310 & -.525675862 & .313713354 & -.197186242 & .047691230 \\ .115589351 & -.404383337 & -.005713670 & -.525675862 & .647636604 & .110185472 & -.079262336 & .000209719 \\ .315461324 & -.102525629 & .441714043 & .313713354 & .110185472 & .546819793 & .306432224 & -.160736733 \\ -.262136144 & .105847039 & -.342388859 & -.197186242 & -.079262336 & .306432224 & .721476885 & .110205575 \\ .126120354 & -.068076824 & .160943854 & .047691230 & .000209719 & -.160736733 & .110205575 & .918313789 \end{pmatrix}.$$

3 A certified exact square root

On $\text{Sym}_8(\mathbb{R})$ use the entrywise norm $\|H\|_{\max} = \max_{i,j} |H_{ij}|$. Let

$$\mathcal{L}_Q(H) = QH + HQ, \quad E = Q^2 - P.$$

Lemma 1. *The operator $\mathcal{L}_Q$ is invertible. There is a symmetric H with*

$$\|H\|_{\max} < 3.014 \cdot 10^{-12}, \quad (Q + H)^2 = P.$$

Proof. The exact checker represents $\mathcal{L}_Q$ on the 36 upper-triangular coordinates. From a rational approximate inverse T it verifies

$$\|I - T\mathcal{L}_Q\|_{\infty} < 1.827 \cdot 10^{-9}, \quad \|\mathcal{L}_Q^{-1}\|_{\infty} < 1192.$$

This is a standard Neumann-series certificate and in particular proves invertibility. The checker also obtains

$$\rho := \|E\|_{\max} < 1.265 \cdot 10^{-15}.$$

Let β denote the exact rational inverse bound produced by this certificate (so $\beta < 1192$), and let $r = 2\beta\rho$. On the closed radius- r ball in $\text{Sym}_8(\mathbb{R})$ consider

$$\Phi(H) = -\mathcal{L}_Q^{-1}(E + H^2).$$

For matrices in that ball,

$$\|H^2 - G^2\|_{\max} \leq 16r\|H - G\|_{\max}.$$

The exact certificate checks

$$q := 4 \cdot 8 \|\mathcal{L}_Q^{-1}\|_{\infty}^2 \rho < 5.748 \cdot 10^{-8} < 1, \quad r < 3.014 \cdot 10^{-12}.$$

Consequently Φ maps the ball to itself and has Lipschitz constant at most q . Banach's fixed-point theorem supplies $H = \Phi(H)$, which is equivalent to $(Q + H)^2 = P$. $\square$

Set $Y = Q + H$. We now have the exact identity

$$I - X^2 - Y^2 = Z^T Z \succeq 0. \tag{2}$$

Thus $(X, Y) \in \mathcal{D}_A(8)$.

4 The matrix-extreme certificate

The columns of $K = K_0/42$, where

$$K_0 = \begin{pmatrix} 39 & 34 & 58 & 31 & 62 \\ -108 & -92 & -236 & 10 & -190 \\ -129 & -64 & -208 & 17 & -176 \\ 42 & 0 & 0 & 0 & 0 \\ 0 & 42 & 0 & 0 & 0 \\ 0 & 0 & 42 & 0 & 0 \\ 0 & 0 & 0 & 42 & 0 \\ 0 & 0 & 0 & 0 & 42 \end{pmatrix},$$

form a basis of $\ker Z$. Indeed $ZK = 0$, and a 3×3 minor of the integer numerator R has determinant -42 , so $\text{rank } Z = 3$. Equation (2) therefore gives

$$\dim \ker L_A(X, Y) = \dim \ker(Z^T Z) = 5,$$

with kernel basis

$$\mathcal{K}(Y) = \begin{pmatrix} K \\ -XK \\ -YK \end{pmatrix}.$$

We use Theorem 2.6(3) of the source paper. It says that (X, Y) is matrix extreme exactly when the only symmetric triple (B_0, B_1, B_2) satisfying

$$B_0v - B_1Xv - B_2Yv = 0, \tag{3}$$

$$B_1v - B_0Xv = 0, \tag{4}$$

$$B_2v - B_0Yv = 0 \tag{5}$$

for every $v \in \ker Z$, together with

$$\text{tr}(B_0 + XB_1 + YB_2) = 0, \tag{6}$$

is the zero triple. In upper-triangular coordinates this is a 121×108 homogeneous linear system; denote its coefficient matrix by $C(Y)$.

The checker selects a fixed 108×108 row minor $M(Y)$ of $C(Y)$. All selected row indices are listed in the code. For the rational center Q , a rational approximate inverse gives the exact Neumann certificate

$$\|I - T_M M(Q)\|_\infty < 2.075 \cdot 10^{-6}, \quad \|M(Q)^{-1}\|_\infty < 3146.$$

Only the terms B_2Yv , B_0Yv , and the YB_2 trace term vary with Y . Since the largest ℓ^1 norm among the columns of K_0 is 544,

$$\|M(Y) - M(Q)\|_\infty \leq c\|Y - Q\|_{\max}, \quad c = \max \left\{ \frac{8 \cdot 544}{42}, 8^2 \right\} = \frac{2176}{21} < 104.$$

The exact certificate finally verifies

$$\|M(Q)^{-1}\|_\infty \|M(Y) - M(Q)\|_\infty < 9.824 \cdot 10^{-7} < 1.$$

Therefore $M(Y)$ is invertible. Hence $C(Y)$ has full column rank, and Theorem 2.6(3) proves that (X, Y) is matrix extreme.

5 Why the point is not free extreme

The source paper's Arveson equation, Theorem 2.6(1), has $gn = 2 \cdot 8 = 16$ real unknowns. Its codomain has dimension

$$d \dim \ker L_A(X, Y) = 3 \cdot 5 = 15.$$

It therefore has a nonzero solution by rank-nullity. Thus (X, Y) is not Arveson extreme. A matrix extreme point is irreducible, and the source paper's Proposition 2.4 and Corollary 2.5 identify free extreme points with irreducible Arveson extreme points. Hence our matrix extreme point is not free extreme. This completes the proof of the theorem.

6 What is exact, and what the computation does

No semidefinite or rank conclusion is inferred from a floating-point tolerance. The checker performs the following proof steps with Python integers and `Fraction` objects:

1. verifies $ZK = 0$, $\text{rank } Z = 3$, and all displayed rational data;
2. constructs the 36-dimensional Lyapunov matrix and the 121×108 matrix-extreme system from their definitions;
3. uses floating point only to propose rational approximate inverses;
4. recomputes both inverse residuals exactly and proves the stated Neumann bounds;
5. proves the contraction and rank-stability inequalities by exact rational comparison.

The two approximate inverse numerators are fingerprinted in the output, but their correctness is not assumed: every run rebuilds and checks them.

7 Search scope, novelty, and review status

The January 24, 2024 author manuscript included here still states that the $g = 2$ case is unknown and makes the wild-disc conjecture. A bounded search on 13 August 2026 for the arXiv identifier, “wild disc”, “non-free matrix extreme”, and later work citing the paper found no subsequent exact two-variable example. This is evidence, not a substitute for a specialist citation-database search.

The result should therefore be treated as a *candidate full solution, likely valid, novelty cautious*. Human review should prioritize (i) the translation from Theorem 2.6 to equations (3)–(6), (ii) the max-norm perturbation constant $2176/21$, and (iii) an independent run of the exact checker. The safety margins are large: the final rank-stability product is below 10^{-6} .

References

- [1] A. Epperly, E. Evert, J. W. Helton, and I. Klep, *Matrix and free extreme points of free spectrahedra*, arXiv:2212.00748; author manuscript dated January 24, 2024.
- [2] E. Evert, J. W. Helton, I. Klep, and S. McCullough, *Extreme points of matrix convex sets, free spectrahedra and dilation theory*, J. Geom. Anal. 28 (2018), 1373–1408.